

The Attention Map as a Lattice Operator: Correlation Length from a Gauge-Equivariant Network’s Gaze

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Abstract

A gauge-equivariant attention network computes its attention weights from a *gauge-invariant* score. This report takes that observation seriously and follows it to its consequence: any reduction of an attention row to a single number per site defines a local gauge-invariant scalar field, i.e. an ordinary lattice operator, and if the network acts on one timeslice at a time then that operator’s zero-momentum connected correlator has an exact transfer-matrix decomposition. The attention field therefore has a mass, and that mass is the theory’s mass gap. We test this in three-dimensional $\mathbb{Z}_2$ gauge theory near its second-order point, where the correlation length is tunable over $\xi = 2.0 \rightarrow 5.6$ lattice spacings, using five ensembles ($24^2 \times 48$, $N = 2000$) and five per-timeslice variational operators trained on a Rayleigh-quotient loss. On 1200 configurations no checkpoint has seen, the correlation length read out of the attention tracks the classical one at **Pearson** 0.9966 over a factor 2.7 of dynamic range, at 5–7% precision on both sides, and — the strongest part of the evidence — it reproduces a *non-monotonic* excursion, $\xi(0.7585) > \xi(0.760)$, measured on the same configurations by two independent operators. A cross- β control matrix shows ξ_A is $11.9\times$ more sensitive to the evaluation ensemble than to the training β , so nothing was memorised, and an explicit configuration-scramble null returns no mass at all.

The result splits into two claims of different strength, and we are careful to keep them apart. The *structural* claim — equivariant attention maps are bona fide lattice operators, with a mass, an overlap, and a spectral decomposition — is established by $\xi_A \approx \xi$, and holds already at random initialisation. The *learning* claim rests instead on the ground-state overlap A_0 : training raises it by $+0.11$ to $+0.27$ (5.4σ to 23.5σ , correlated differences on shared configurations) and raises the site-to-site fluctuation of the attention by one to one and a half orders of magnitude. “The network discovers the correlation length” is *not* a supportable statement; “training removes excited-state contamination from the attention field” is. We also record why two earlier attempts at the same question failed — both measured the mean attention *kernel*, a statistic bounded by the receptive radius R and centred by the ball geometry, rather than the attention *field* — and the four analysis defects that had to be repaired before the numbers above could be trusted, the largest being a generalized eigenproblem that was selecting a near-null direction at every β .

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1 The question, and why it is not a trivial one

The interpretability program of this thesis rests on one premise: in a gauge-equivariant network, the attention map is not a diagnostic heat map but a *measurement*. The conference abstract commits to two investigations of that premise — whether attention localizes in topologically rich regions, and whether the **attention range correlates with the physical correlation length**. This report is about the second.

Stated naively the question is almost too easy to fail: train one operator per inverse coupling β , measure how far each attention head looks, plot it against the correlation length $\xi(\beta)$, and see whether the two rise together. That programme was run twice and produced a negative both times (Section 3). The diagnosis — that the failures were a property of the *statistic* and not of the physics — is what makes the rest of the report possible, so it is worth stating the trap early: a range statistic computed from the *lattice-averaged* attention kernel is bounded above by the receptive radius R and centred by the geometry of the offset ball, and it discards precisely the site-to-site variation that distinguishes attention from convolution.

The replacement statistic is the subject of Section 2.4. Instead of asking how *wide* the attention is, **we treat the attention itself as a field on the lattice and ask what *mass* it has**. That question has a sharp, falsifiable answer supplied by the transfer matrix, needs no reference to the receptive field at all, and — unlike the range — does not exist for a convolution.

2 Theoretical background

2.1 Masses, correlators, and the mass gap

Euclidean lattice field theory computes expectation values as statistical averages over gauge-field configurations U with weight $e^{-S[U]}$. For a lattice with a periodic temporal extent N_t and a transfer matrix $\hat{T} = e^{-\hat{H}}$, the correlator of a zero-momentum operator

$$\bar{O}(t) = \sum_{\vec{x}} O(t, \vec{x}) \tag{1}$$

admits the spectral decomposition

$$C(\Delta) = \langle \delta \bar{O}(t + \Delta) \delta \bar{O}(t) \rangle = \sum_{n>0} |\langle 0 | \hat{O} | n \rangle|^2 \left(e^{-m_n \Delta} + e^{-m_n(N_t - \Delta)} \right), \quad (2)$$

where $\delta \bar{O} = \bar{O} - \langle \bar{O} \rangle$ (the vacuum subtraction is mandatory for any operator with the vacuum quantum numbers, which is every operator considered here) and the second exponential is the periodic image. At large Δ the lightest state dominates, so

$$C(\Delta) \xrightarrow{\Delta \text{ large}} A \left(e^{-m_0 \Delta} + e^{-m_0(N_t - \Delta)} \right), \quad \xi = \frac{1}{m_0}. \quad (3)$$

Two derived quantities are used throughout:

- the **effective mass** $m_{\text{eff}}(\Delta) = \log[C(\Delta)/C(\Delta + 1)]$, which converges to m_0 from above — every excited state in Eq. (2) adds a faster-decaying exponential, so a contaminated correlator decays too fast and reads *m high*, hence ξ *low*. This sign matters in Section 6.2;
- the **ground-state overlap fraction**

$$A_0 = \frac{A(1 + e^{-m_0 N_t})}{C(0)} \in [0, 1], \quad (4)$$

the fraction of the operator’s own variance carried by the ground state. This is Morningstar–Peardon’s measure of operator quality, and the quantity a variational construction maximises. Two operators measuring the same physics must agree on m_0 ; they may differ arbitrarily on A_0 , and that is where operator quality lives.

2.2 The testbed: 3D $\mathbb{Z}_2$ gauge theory

The measurement needs a correlation length that is (i) large compared with one lattice spacing, so that a *length* is resolvable at all on an integer grid, and (ii) tunable over a real dynamic range, so that a correlation between two lengths is more than a two-point coincidence. Three-dimensional $\mathbb{Z}_2$ gauge theory delivers both: it is dual to the 3D Ising model and has a second-order transition at $\beta_c \approx 0.7614$, where the mass gap closes and $\xi = 1/m$ diverges as $\xi \sim (\beta_c - \beta)^{-\nu}$.

This is a deliberate change of system. The $SU(2)$ glueball ensembles on which the rest of this thesis’s spectroscopy is done have $\xi_s \leq 1.1$ spatial lattice spacings at every β examined (Section 3), which makes the question unaskable there for reasons that have nothing to do with attention. $\mathbb{Z}_2$ is also the project’s declared debug testbed ($n_c = 1$, links ± 1) and is cheap. The price is that every conclusion below is a statement about the *method*, not about $SU(N)$.

Five ensembles were generated at

$$\beta \in \{0.7450, 0.7520, 0.7560, 0.7585, 0.7600\}, \quad 24^2 \times 48, \quad N = 2000, \quad (5)$$

spanning $\xi \approx 2.0 \rightarrow 5.6$. Their generation required an exact $\mathbb{Z}_2$ heat-bath, $P(U = +1) = \sigma(2\beta s)$ with s the staple sum: the single-link Metropolis proposal is the flip $U \rightarrow -U$, whose acceptance *collapses* to ≈ 0.02 as the configuration orders near β_c , freezing the chain exactly where ξ is largest. Even with the heat-bath, critical slowing down ($\tau \sim \xi^z$, $z \approx 2$) drives the integrated autocorrelation time to $\tau_{\text{int}} \approx 44$ sweeps, which fixes the production $n_{\text{skip}} = 200$.

2.3 Gauge-equivariant attention, and why its weights are physical

The GELT block computes, for each query site x , a score against every site $x + \Delta$ in the L_1 ball of Manhattan radius R . The key and value matrices at $x + \Delta$ are first *parallel-transported* to x along the (averaged) shortest lattice paths, and the score is the real trace

$$s_{x \rightarrow x+\Delta}^{(\ell, h)} = \text{Re Tr} [Q^\dagger(x) \tilde{K}(x+\Delta)], \quad \tilde{K}(x+\Delta) = T_\Delta(x) K(x+\Delta) T_\Delta^\dagger(x), \quad (6)$$

followed by a softmax over Δ giving the attention weights $\alpha_{x \rightarrow x+\Delta}^{(\ell, h)}$. Under a local gauge transformation $\Omega(x)$, both $Q(x)$ and the transported $\tilde{K}$ conjugate by the *same* $\Omega(x)$, so the trace in Eq. (6) is **invariant**, and therefore so is α .

This is the entire foundation of what follows, and it is a property of the architecture rather than of the training. It is also exactly what a non-equivariant network does not have: there, the attention weights depend on the gauge in which the configuration happens to be written, and no gauge-fixing convention makes a gauge-dependent quantity into an observable.

2.4 The claim: the attention field has a mass

Because $\alpha_{x \rightarrow \cdot}$ is gauge invariant at every site, *any* reduction of the attention row to a scalar,

$$A(x) = f(\alpha_{x \rightarrow \cdot}), \quad (7)$$

is a gauge-invariant local scalar field built from the links — a lattice operator in the ordinary sense, with all the machinery of Section 2.1 available to it. **Three reductions are used here**, chosen to be qualitatively different rather than exhaustive:

$$\underbrace{\alpha_{x \rightarrow x}}_{\text{self}}, \quad \underbrace{\sum_{\Delta} |\Delta|_1 \alpha_{x \rightarrow x+\Delta}}_{\text{ell}}, \quad \underbrace{- \sum_{\Delta} \alpha_{x \rightarrow x+\Delta} \log \alpha_{x \rightarrow x+\Delta}}_{\text{ent}}. \quad (8)$$

Note that **ell** is the *local, per-site* version of the failed range statistic: the trap of Section 3 is not the reduction, it is the averaging that used to follow it.

The trained operators are **per-timeslice** by construction — the transfer-matrix bound that makes the Rayleigh loss a mass estimate requires $\bar{O}(t)$ to depend on slice t alone, so the network sees one timeslice at a time and has no temporal receptive field. Consequently $A(t, \vec{x})$ depends on slice t alone, its zero-momentum sum $\bar{A}(t) = \sum_{\vec{x}} A(t, \vec{x})$ is a legitimate timeslice operator, and Eq. (2) applies verbatim:

$$C_A(\Delta) = \langle \delta \bar{A}(t + \Delta) \delta \bar{A}(t) \rangle = \sum_{n>0} |\langle 0 | \hat{A} | n \rangle|^2 e^{-m_n \Delta} + \dots \quad (9)$$

So the attention field has a mass m_A , and

$$\boxed{\xi_A = 1/m_A \stackrel{?}{=} \xi(\beta)} \quad (10)$$

with *no ceiling at R* and *no geometric offset*, because Eq. (9) correlates a fluctuation — zero-mean by construction — and not a moment of a kernel.

Three consequences, in increasing order of interest:

1. ξ_A **tracks** $\xi(\beta)$: the length the failed statistic was supposed to find, recovered from the fluctuations of the attention rather than from its width.
2. **This measurement does not exist for a convolution.** A fixed kernel has $\delta A \equiv 0$ identically, so $C_A \equiv 0$. The mere *existence* of a signal is the content-dependence result, and it is quantitative: the site-level fluctuation-to-mean ratio $\delta A/A$ is its effect size.
3. **A_0 of the attention field** says whether *training* taught the routing to look at the physics, or whether the equivariant construction alone is responsible.

2.5 Why this is a measurement and not a tautology

Point 1 above is, on its own, close to a theorem: *any* local gauge-invariant scalar has a correlator that decays at the mass gap, a random-init network’s attention field included. Finding $\xi_A \approx \xi$ therefore establishes point 2 — that attention maps in an equivariant network are physical fields — but *says nothing whatsoever about learning*. The design must separate the two, and does so with three arms:

arm	what it isolates
trained network, matched β (diagonal)	the headline ξ_A vs ξ
trained network, mismatched β (off-diagonal)	content-dependence vs memorisation: ξ_A must follow the <i>evaluation</i> ensemble, not the training β
random-init network, all β	what equivariance gives <i>before</i> learning: the A_0 baseline

The off-diagonal arm also removes the confound that killed the previous attempt (Section 3): when one network per β is read out on its own ensemble, operator quality varies along the very axis being plotted. With one fixed network evaluated across five ensembles, operator quality is a property of the *row* and the physics being probed is a property of the *column*.

3 Two prior negatives, and their shared cause

3.1 $SU(2)$: the question is unaskable

A classical GEVP β -scan on the $12^3 \times 24$, $\xi = 3$ anisotropic $SU(2)$ family (the same lattice on which this thesis’s glueball result was obtained) gives spatial correlation lengths

β	2.1	2.3	2.4	2.5	2.7
$\xi_s [a_s]$	0.57	0.83	1.00	1.10	1.00

ξ_s never exceeds 1.1 lattice spacings. Attention offsets are integers, so the entire variation a range statistic would have to track lives inside the first offset shell. The box is not the problem — $L/\xi_s \approx 12$, the regime in which finite-volume effects are *absent* — the spacing is: at $\xi = 3$ the spatial coupling $\beta_s = \beta/\xi \approx 0.8$ is strong coupling, a_s is coarse, and the whole glueball fits inside one grid cell.

3.2 $\mathbb{Z}_2$: real dynamic range, and the statistic still fails

Moving to 3D $\mathbb{Z}_2$ bought $\xi = 2.05 \rightarrow 5.28$, i.e. genuine range. One variational operator per β was trained, and the mean attention range

$$\ell_{\text{att}} = \sum_{\Delta} |\Delta|_1 \bar{\alpha}(\Delta), \quad \bar{\alpha} = \text{attention averaged over sites and configurations} \quad (11)$$

was read out. At $R = 6$ it gave 4.25, 4.54; at $R = 12$ it gave 8.63, 7.90, 8.13, 7.44, 8.21 — scatter, no trend in ξ .

The reason is that ℓ_{att} has a **uniform reference** that nobody had quoted. For attention spread uniformly over the 2D L_1 ball,

$$\ell_{\text{uniform}} = \frac{\sum_d 4d \cdot d}{n_{\text{off}}} = 4.28 \quad (R = 6), \quad 8.31 \quad (R = 12), \quad (12)$$

and *every measurement landed on that number*. The statistic is bounded above by R and centred by the ball geometry; only the bound had been guarded against. In hindsight the $R = 6$ “trend”

($4.25 \rightarrow 4.54$ about a uniform 4.28) was near-uniform noise and should never have been read as a signal. A later reproducibility check settled it independently: retraining at the same R and the same β with more data moved ℓ_{att} by 0.3–0.5, against an across- β span of $3.95 \rightarrow 4.57$ for the whole scan. The run-to-run variation of the statistic is the size of the entire effect it was supposed to measure.

3.3 The diagnosis

Both failures share one cause. $\bar{\alpha}$ is an average *over the lattice*, so ℓ_{att} is a property of the learned *kernel* — a design quantity of the same kind as a convolution’s filter width — and averaging is exactly the operation that discards the site-to-site, configuration-to-configuration variation that attention has and convolution does not.

That the discarded part is large was already visible in the $R = 12$ readout: the per-head offset entropy relative to its uniform ceiling at $\beta = 0.745$ reads 0.87, 0.92, 0.93, 0.97, **0.42**, 0.73, 0.97, **0.72**. Several heads are strongly structured — while their ℓ_{att} sits at 8.3 against a uniform 8.31. The attention was doing something; the radial first moment was blind to it. **Keeping the variation instead of averaging it away is the whole content of Section 2.4.**

4 What we did

4.1 Ensembles and trained operators

Everything the measurement needs already existed when it was designed; no sampling and no training were required, only forward passes.

Ensembles. Five $\mathbb{Z}_2$ ensembles, $24^2 \times 48$, $N = 2000$ configurations each, exact heat-bath, $n_{\text{skip}} = 200$ (Section 2.2).

Operators. One GELT variational operator per β , trained exactly as the $SU(2)$ glueball operator is: GELT with 4 GEMHSA layers, 2 heads, $d_{\text{model}} = 16$, $d_{qkv} = 8$, RoPE positional encoding, receptive radius $R = 6$, fed four cumulative APE-smearing levels (0, 4, 8, 16) as input channels, and trained on the Rayleigh loss $-\overline{C(\Delta)}/\overline{C(0)}$ over $\Delta \in \{1, 2\}$ whose converged value *is* (minus the exponential of) the mass. The operator is per-timeslice — 2D here, since 3D $\mathbb{Z}_2$ has two spatial directions — so the transfer-matrix bound of Section 2.4 holds.

Training used `configs[:800]` at every β . **All measurements below use `configs[800:]`**, so the evaluation slice is unseen by every checkpoint, the diagonal included.

Operator quality. $R = 6$ with 800 training configurations was chosen after an $R = 12$ sweep produced systematically undertrained operators (halving the data to pay for 313 offsets left 70 optimizer steps/epoch instead of 140). Each checkpoint is gated against the classical mass with *matched estimators* — the network’s own correlator re-fitted with the classical scan’s window and blocked jackknife, with both errors propagated — because the first version of the gate compared $m_{\text{eff}}(\Delta=1)$ against a cosh fit and thereby manufactured an apparent “undertraining that grows with ξ ” out of a purely one-sided estimator bias. All five $R = 6$ checkpoints pass the corrected gate.

4.2 The measurement pipeline

For each (network, ensemble) pair, and for each of the $4 \text{ layers} \times 3 \text{ reductions} \times 2 \text{ heads} = 24$ attention channels:

1. run the forward pass on the unseen configurations and stash `_last_alpha` per layer (under `no_grad`);
2. reduce each attention row to a scalar per site via Eq. (7), and accumulate the site-level first and second moments so the fluctuation-to-mean ratio $\delta A/A$ can be quoted;
3. form the zero-momentum series $\bar{A}(t)$ per configuration;
4. project the multi-channel basis onto its GEVP ground-state vector ($t_0 = 1$, $t_d = 2$), with the safeguards of Section 6.3;
5. fit the projected connected correlator to the periodic single-state form over $\Delta \in [2, 8]$;
6. take errors from a **blocked** jackknife over configurations with block 20.

Conventions in steps 4–6 are copied verbatim from the classical β -scan so the two masses are directly comparable, and the classical smeared-plaquette basis (levels 0, 4, 8, 16) is **re-measured on the same configurations**, so $\xi_A - \xi_{\text{class}}$ can be quoted as a correlated difference rather than as two independent numbers.

Reference arms. Alongside the attention channels we measure the network’s own output operator $\bar{O}(t)$ (a check that the checkpoint loaded correctly: its mass must reproduce the recorded one) and the classical basis.

Nulls. Two, and the distinction between them turned out to matter (Section 6.3):

- **config-scramble** — each timeslice is drawn from an independently permuted configuration. This destroys the temporal correlation *and* the per-configuration zero mode, so $C(\Delta > 0)$ has expectation zero and there is nothing for a mass to be fitted to. This is the null proper.
- **time-shuffle** — one independent permutation of the timeslices per configuration. This is *not* a null (it preserves each configuration’s mean over t); it is retained as a **zero-mode consistency check**, whose plateau must equal $(2\xi - 1)/N_t$.

Cost. Two runs. The *diagonal* run (matched network + random control, $N_{\text{eval}} = 1200$ configurations per ensemble) buys precision where precision is the point. The *cross* run (all five networks + random on all five ensembles, $N_{\text{eval}} = 600$) buys the control matrix, where only the row/column contrast matters. About four minutes per ensemble per network.

5 Results

5.1 The headline: the attention field carries the mass gap

Table 1 is the diagonal run: every trained network on its own ensemble, on the 1200 configurations it never saw, against the classical basis measured on those same configurations.

$$\text{Pearson}(\xi_A, \xi_{\text{classical}}) = \mathbf{0.9966}, \quad \text{slope } 1.21, \quad (13)$$

over a factor 2.74 of classical dynamic range, at 5–7% precision on both sides. The claim of Section 2.4 holds: the attention map is a lattice operator, and its correlator decays with the mass gap.

Table 1: Correlation lengths from the attention field, from the classical smeared basis, and from a randomly initialised network of identical architecture. 1200 unseen configurations per ensemble; errors are blocked jackknife (block 20). “channel” is the attention channel the variational self-check selected (layer, head, reduction). ξ_{out} is the network’s own output operator, shown for reference.

β	ξ classical	ξ_A trained	channel	ξ_A random init	ξ_{out}
0.7450	2.04 (14)	2.24 (12)	L4h0:sel f	2.08 (13)	2.33
0.7520	2.34 (16)	2.76 (13)	L4h1:sel f	2.46 (16)	2.92
0.7560	4.08 (27)	4.48 (30)	L4h0:e ll	4.08 (26)	4.63
0.7585	5.59 (34)	6.64 (32)	L4h0:ent	5.47 (31)	6.65
0.7600	4.71 (34)	5.60 (37)	L4h1:ent	4.80 (34)	5.81

The correlation survives a non-monotonic excursion. This is the strongest single piece of evidence in the report and it deserves to be read carefully. $\xi(0.7585) > \xi(0.760)$ — i.e. the point *further* from β_c has the *larger* correlation length — in the classical measurement *and* in the attention, on the same configurations, from independent operators. Two quantities that both rise with β correlate for trivial reasons; two that both turn around at the same place do not. (An earlier $N = 2000$ scan had the opposite ordering at these two points; those were its two worst-determined points, measured with the defective GEVP of Section 6.3, and they are superseded.) The excursion is a property of the *ensembles*, not of the attention: both points sit at $\xi \gtrsim L/4 = 6$, the scan’s own finite-volume guard — see Section 6.4.

Correlator profiles. At $\beta = 0.760$, the point closest to β_c , $C(\Delta)/C(0)$ reads

	$\Delta = 0$	1	2	3	4	5	6
classical smeared basis	+1.000	+0.404	+0.272	+0.197	+0.163	+0.134	+0.109
trained attention	+1.000	+0.619	+0.470	+0.373	+0.312	+0.260	+0.224
random attention	+1.000	+0.466	+0.310	+0.226	+0.186	+0.155	+0.128
config-scramble null	+1.000	+0.005	+0.007	+0.009	+0.010	−0.007	−0.000

The trained attention field is a visibly cleaner exponential than either the classical basis or the untrained network, and the null is flat at the 10^{-2} level with no fittable mass.

5.2 Training separates from initialisation at every β

Table 2 is the learning claim. ΔA_0 is a blocked jackknife of the *difference* on shared configurations: both arms see identical configurations, so the naive $\sqrt{\sigma^2 + \sigma^2}$ throws away the cancellation.

Table 2: Ground-state overlap of the attention field, trained vs randomly initialised, and the site-level fluctuation-to-mean ratio $\delta A/A$ (median over the 24 channels). ΔA_0 and $\Delta \xi$ are correlated differences on shared configurations.

β	A_0 trained	A_0 random	A_0 classical	$\delta A/A$ (tr / rnd)	ΔA_0 (correlated)	$\Delta \xi$
0.7450	0.848 (41)	0.704 (42)	0.578 (39)	0.091 / 0.0067 ($\times 14$)	$+0.144 \pm 0.017$ (8.4 σ)	$+0.17 \pm 0.04$
0.7520	0.782 (28)	0.671 (36)	0.576 (37)	0.140 / 0.0060 ($\times 23$)	$+0.110 \pm 0.020$ (5.4 σ)	$+0.31 \pm 0.07$
0.7560	0.738 (17)	0.534 (16)	0.449 (15)	0.246 / 0.0054 ($\times 46$)	$+0.204 \pm 0.012$ (16.9 σ)	$+0.41 \pm 0.13$
0.7585	0.753 (11)	0.486 (14)	0.413 (13)	0.113 / 0.0049 ($\times 23$)	$+0.267 \pm 0.011$ (23.5 σ)	$+1.17 \pm 0.19$
0.7600	0.654 (14)	0.449 (15)	0.397 (14)	0.130 / 0.0046 ($\times 28$)	$+0.205 \pm 0.012$ (16.9 σ)	$+0.80 \pm 0.19$

Three things to read off it. First, the trained attention field is a *better operator than the classical smeared basis* at every β , by 0.20 to 0.34 in overlap fraction — and the classical basis

degrades towards β_c ($0.578 \rightarrow 0.397$) faster than the attention does ($0.848 \rightarrow 0.654$). Second, the random network’s site-to-site fluctuation is ≈ 0.005 essentially independently of β , and training raises it by one to one and a half orders of magnitude; for a fixed convolutional kernel the same quantity is identically zero. Third, $\Delta\xi > 0$ everywhere: the trained attention field is also *less* contaminated by excited states than the untrained one, not merely better normalised.

Autocorrelation. τ_{int} of the selected channel along the chain reads 0.63, 1.13, 2.65, **8.35**, 5.85 across the five β . The jackknife block of 20 clears $2\tau_{\text{int}}$ at every β , but only just at $\beta = 0.7585$.

The zero-mode check. The time-shuffled arm’s plateau matches its prediction $(2\xi - 1)/N_t$ at every β : measured 0.062, 0.079, 0.138, 0.231, 0.159 against predicted 0.073, 0.094, 0.166, 0.256, 0.212, tracking at Pearson 0.98 *including* the non-monotonic excursion. This is a second, independent confirmation of ξ from the same data.

5.3 The cross- β control matrix: no memorisation

Five trained networks plus the random arm, each evaluated on all five ensembles, 600 unseen configurations per cell. One cell (**train@0.756** evaluated on $\beta = 0.752$) returned $A_0 = 5.76$ with $\xi = 0.39$ — a fit that ran to its grid edge — and is masked throughout; the guard now reports such cells as unresolved rather than as numbers.

Table 3: ξ_A from the cross-evaluation matrix. Rows are the β at which the network was trained; columns are the ensemble it is evaluated on. The masked cell is marked —.

network	evaluation ensemble β				
trained at	0.7450	0.7520	0.7560	0.7585	0.7600
0.7450	2.24	2.73	4.53	5.73	4.85
0.7520	2.33	2.77	4.64	5.74	4.92
0.7560	2.36	—	4.71	6.06	5.04
0.7585	2.31	2.80	4.68	6.50	5.25
0.7600	2.22	2.69	4.68	6.15	5.13
random init	2.05	2.47	4.32	5.42	4.77
<i>classical</i>	2.05	2.36	4.21	5.48	4.73

The matrix reads across, not down:

$$\frac{\text{spread within a row (across evaluation ensembles)}}{\text{spread within a column (across training } \beta)} = \frac{1.405}{0.118} = \mathbf{11.9}, \quad (14)$$

and the column means correlate with the classical ξ at **Pearson** 0.9983. The attention reads the configuration in front of it, and a fixed kernel has no analogue of this measurement at all.

The transfer test. If a network had learned its own ensemble rather than physics, the diagonal would dominate its column. It does not:

β	A_0 diagonal	A_0 off-diagonal	advantage	A_0 random
0.7450	0.855	0.770	+0.085	0.723
0.7520	0.774	0.815	− 0.041	0.656
0.7560	0.697	0.687	+0.010	0.484
0.7585	0.764	0.696	+0.069	0.495
0.7600	0.660	0.640	+0.020	0.423

The mean diagonal advantage is $+0.029$ and is *inconsistent in sign*, against a mean trained-minus-random of $+0.167$: the learning effect is $6\times$ the β -matching effect. **Every trained network beats the random arm on every ensemble** — the worst trained network on each ensemble scores $0.747/0.774/0.636/0.659/0.571$ against the random arm's $0.723/0.656/0.484/0.495/0.423$.

What the matrix *does* separate is network quality: ensemble-averaged A_0 per row is $0.734/0.677/0.702/0.775/0.726$ against 0.556 for random, so **train@0.7585** is the best operator on three of five ensembles, including ones it never saw — a globally better operator, not a β -specialist. **The learning claim is therefore about transferable physics**: the routing did not memorise a correlation length, it learned something that works at every ξ in the scan.

The null passes. With the config-scramble, 26 of the 30 cells do not fit at all — there is nothing to fit — and the four that do return $|A_0| \leq 0.0066$.

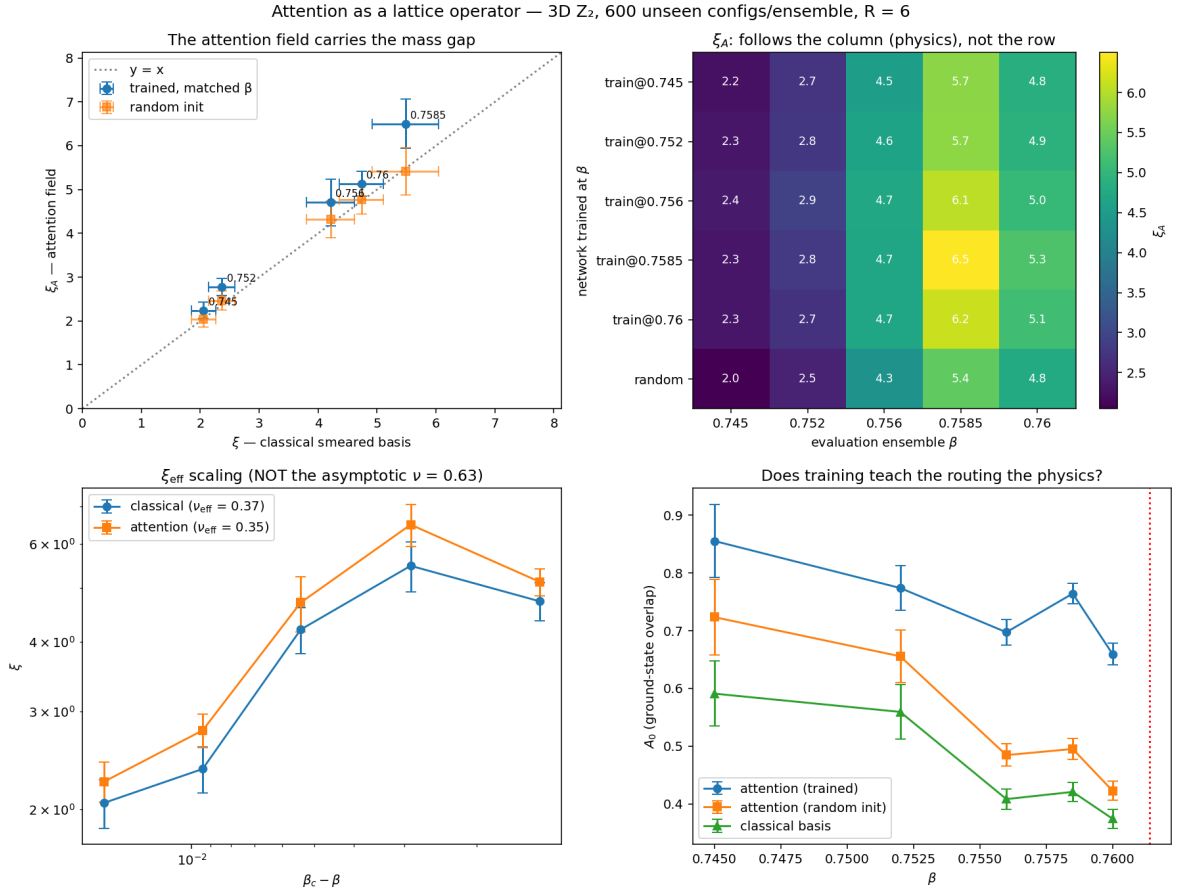


Figure 1: The four panels of the cross-evaluation run. **Top left:** ξ_A against the classical ξ measured on the same configurations, for the matched trained network and for random initialisation; the dotted line is $y = x$. **Top right:** the cross-evaluation matrix of Table 3 — ξ_A follows the column (the physics), not the row (the training β). **Bottom left:** both lengths against $\beta_c - \beta$ with an effective exponent; this is *not* a measurement of ν (Section 6.4). **Bottom right:** ground-state overlap A_0 for the trained attention, the random attention, and the classical basis.

6 Analysis

6.1 What is established, and what is not

The random-init arm *also* tracks ξ , at Pearson 0.9989 — if anything more tightly than the trained arm. This is exactly what Section 2.5 predicted, and it is not a disappointment; it is the correct partition of two claims of very different strength:

- **Structural claim (established).** Attention maps in a gauge-*equivariant* network are bona fide lattice operators, with a mass, an overlap, and a spectral decomposition. This is established by $\xi_A \approx \xi$, and it is true already at initialisation. It is the premise of the whole interpretability program made quantitative, and it is a property of the *architecture*.
- **Learning claim (established, but by a different statistic).** Training makes those operators *good*. This is established by A_0 (+0.11 to +0.27, up to 23.5σ), by $\delta A/A$ ($\times 14$ to $\times 46$), and by the transfer test of Section 5.3 — *not* by ξ_A .

“The network discovers the correlation length” is **not** a supportable claim and must not be made.

A known confound on $\delta A/A$. α is a softmax output, so a network with larger score magnitudes has a sharper — hence more variable — attention regardless of what it learned. $\delta A/A$ is therefore *descriptive*; the load-bearing learning statistic is A_0 , which a sharp-but-untrained attention would not raise (it would build a UV-dominated field with *poor* ground-state overlap). Closing this properly needs an entropy-matched random arm, which has not been run.

6.2 Why the trained ξ_A overshoots, and why “closer to classical” is the wrong target

The natural inference from Section 5.2 is: the trained attention has the better ground-state overlap, so its ξ_A should sit *closer* to the classical ξ . The data say the opposite, and the wrong reading here would invert the conclusion.

β	ξ classical	ξ_A trained	ξ_A random	trained – class	random – class
0.7450	2.04	2.24	2.08	0.20	0.04
0.7520	2.34	2.76	2.46	0.42	0.12
0.7560	4.08	4.48	4.08	0.40	0.00
0.7585	5.59	6.64	5.47	1.05	0.12
0.7600	4.71	5.60	4.80	0.89	0.09

The *random* arm is the one that agrees with the classical operator, at every β (fitted slope 0.96 against the trained arm’s 1.21). This is not a contradiction, because **the classical operator is not ground truth**. It is one more operator with its own excited-state contamination, and contamination has a *signed* effect: by Eq. (2), excited states add faster-decaying exponentials, so a contaminated correlator decays faster than the true gap and reads ξ **low**. A_0 measures how much of an operator’s own $C(\Delta)$ is the ground state, so a higher A_0 means less of that downward bias, and the operator with the highest A_0 should read the *largest* ξ . It does. The observed ordering

$$\xi_{\text{trained}} > \xi_{\text{random}} \approx \xi_{\text{classical}} \quad (15)$$

is precisely the ordering the A_0 column predicts, once one stops treating “classical” as the target. The correlator profiles in Section 5.1 say the same thing directly: at $\beta = 0.760$ the

classical basis has fallen to 0.404 by $\Delta = 1$ and the random attention to 0.466, while the trained attention still holds 0.619.

So the natural reading of the whole result is that the trained operator’s larger ξ is the *candidate* for the value closest to the true gap, and that **learning = removing excited-state contamination** is the physically honest gloss of the A_0 result. That gloss is stronger than the correlation framing, because it says what the training bought in operator terms — which is exactly what a variational-operator audience asks for.

Stated as a candidate, not as a result. There is no independent ground truth for ξ at these β : the older $N = 2000$ scan is superseded by the GEVP defect below, and both top points sit at $\xi \gtrsim L/4$ where finite volume bites. A 21% systematic in the attention correlator fit would look identical. Distinguishing “de-contamination” from “systematic overshoot” needs a reference that is not itself contaminated. Consequently:

- do **not** quote $|\xi_A - \xi_{\text{classical}}|$ as an accuracy metric, and do not present the random arm’s smaller residual as the random arm being *better* — agreement with a contaminated reference is not accuracy;
- the load-bearing statements remain the *correlation* (Pearson 0.9966, through the non-monotonic excursion) for the structural claim and A_0 for the learning claim. The slope 1.21 is a separate, open quantity.

6.3 Four defects, and how each was resolved

None of the numbers above survived the first version of the analysis. The repairs are recorded because two of them are of general interest.

1. The GEVP was selecting a near-null direction — at every β . The first suspicion was our own basis pruning; fixing it moved the classical masses by 4×10^{-15} . That non-result *is* the diagnosis: m , A_0 and $C(\Delta)/C(0)$ are all invariant under rescaling the projected operator, so identical values mean the same direction was selected either way. The real cause was an eigenvalue floor of 10^{-12} relative to the largest in the whitening step — twelve orders of magnitude, i.e. no regularisation at all. A nested-smearing $C(t_0)$ has three or four meaningful orders, and a small signal divided by a floored *noise* eigenvalue produces a spuriously large generalized eigenvalue pointing into noise: a near-cancelling combination whose $C(0)$ is almost pure noise while $C(\Delta > 0)$ still carries the physical decay. The symptom was $C(1)/C(0) = 0.039/0.020/0.007$ at the three largest β with 36% / 87% / unresolved errors — *correct masses, useless error bars*.

Fixed with a floor of 10^{-4} plus a **variational self-check**: v_0 maximises the Rayleigh quotient over the span of the basis, so the projection can never interpolate worse than any single member; when it does, fall back to that member. The decision is taken once on the full sample and held fixed across jackknife replicas (a jackknife whose estimator changes between replicas does not estimate anything). All five β then resolve at 6–7% with $C(1)/C(0) = 0.39\text{--}0.44$ uniformly — and the **fallback fires at every β** , on both the classical and the attention bases.

2. The multi-channel attention GEVP is unstable. Three NaN cells and one absurd one ($\xi = 4.85 \pm 89$) out of ten. Moot after the self-check: it falls back to the single best channel at every β , so the multi-channel and best-single arms coincide and the tables quote one number. Informative in itself — the 24 attention channels are redundant enough that a variational combination never helps.

3. ΔA_0 errors were uncorrelated. Fixed with a blocked jackknife of the difference on shared configurations. The two weakest points went from 2.4σ to 8.4σ and 5.4σ ; nothing else moved.

4. A time-shuffle is not a null. The null’s A_0 rose from 0.005–0.014 to 0.043–0.230 once the GEVP fallback landed. The first suspicion was selection bias (the null re-selecting its own best-of-six; that bias is real, and measures $7.4\times$ on pure noise). Removing it changed the null by 0.000. The correct explanation is structural: a permutation of timeslices preserves each configuration’s mean over t , while the connected correlator subtracts only the *global* mean, so the residual retains

$$\text{Var}_{\text{config}}(\bar{O}_c) = C(0) \cdot \frac{2\xi - 1}{N_t} \quad (16)$$

as a *flat pedestal at every Δ* — which is exactly what the profiles show ($+1.000, +0.236, +0.229, +0.231, +0.234, \dots$ at $\beta = 0.7585$: no decay at all). The “null” was measuring the finite- N_t zero mode — physics, not noise — and could never have gone to zero. Fixed with the config-scramble of Section 4.2, and validated on a synthetic operator of known mass ($\xi = 5.56$ injected): the real arm returns $\xi = 5.65$ with $A_0 = 1.000$, the time-shuffle plateau lands at 0.213 against a predicted 0.215, and the config-scramble returns ± 0.001 with no fittable mass. The time-shuffle is retained as the zero-mode consistency check of Section 5.2.

6.4 Limitations, stated up front

1. **One theory, one volume, one lattice family.** 3D $\mathbb{Z}_2$ at $24^2 \times 48$. This is a statement about the *method*, not about $SU(N)$. The natural next test is the $SU(2)$ glueball operator, where the same readout machinery already exists.
2. **Finite volume at the top two β is the open physics question.** $\xi = 5.6$ and 4.7 against $L/4 = 6$ puts both points at or past the scan’s own guard, and the non-monotonic excursion is what a volume-squeezed pair looks like. The test is $L = 24 \rightarrow 32$ — but GELT is built per- L , so that means new ensembles *and* new checkpoints at all five β ($\approx 2.4\times$ per sweep, tens of hours), not one point. Worth it only if the excursion is to be reported as physics rather than flagged as unresolved. It is flagged as unresolved here.
3. **The effective exponent is not ν .** Fitting $\xi \sim (\beta_c - \beta)^{-\nu}$ to these five points returns 0.35–0.48 depending on the arm and the configuration budget (0.43 classical and 0.48 attention on the diagonal run, 0.37 and 0.35 on the cross run of Figure 1), against the 3D Ising $\nu = 0.63$: $\xi \leq 5.6$ at $L = 24$ is not the asymptotic scaling regime, and the non-monotonic excursion sits inside the fit. It may be quoted only as a *consistency* check between the attention and the classical scan on the same points — and with the excursion now confirmed on both operators, it should probably not be quoted at all until the volume question is settled.
4. **Operator quality is uneven across the five checkpoints.** The cross-evaluation design makes this a nuisance parameter rather than a confound (quality is a property of the row, physics of the column), but the diagonal alone would still inherit it.
5. **The $\delta A/A$ confound** of Section 6.1: no entropy-matched random arm has been run.
6. **One masked cell** in the cross matrix (Section 5.3), excluded by an automatic guard rather than by hand, but excluded nonetheless.

7 Conclusions

The question the conference abstract poses — does the attention range track the physical correlation length? — was closed negative twice, and both negatives were about the statistic rather than the physics. ℓ_{att} , the first radial moment of the *lattice-averaged* attention, is bounded above by the receptive radius and centred by the geometry of the offset ball; it is a property of the learned kernel, of the same kind as a convolution’s filter width, and the averaging that defines it discards exactly what distinguishes attention from convolution. **Any range statistic must be quoted against its uniform reference**, and this one never survives that comparison.

Keeping the site-to-site variation instead turns the question into a well-posed one. Because the equivariant score is gauge invariant, any reduction of an attention row is a local gauge-invariant scalar field; because the trained operator is per-timeslice, that field’s zero-momentum correlator has an exact transfer-matrix decomposition. The attention field therefore *has a mass*, and it is the theory’s mass gap. Measured on 1200 unseen configurations per ensemble across five 3D $\mathbb{Z}_2$ ensembles:

- ξ_A tracks the classical ξ at **Pearson** 0.9966 over $\times 2.7$ of dynamic range at 5–7% precision on both sides, and reproduces a non-monotonic excursion in β that two independent operators see on the same configurations;
- ξ_A is **11.9** $\times$ more sensitive to the evaluation ensemble than to the training β , with column means correlating with the classical ξ at Pearson 0.9983, so the networks read the configuration in front of them rather than recalling their own;
- a config-scramble null returns no mass in 26 of 30 cells and $|A_0| \leq 0.0066$ in the rest;
- training raises the ground-state overlap of the attention field by +0.11 to +0.27 (5.4σ to 23.5σ) over random initialisation, and raises its site-level fluctuation by one to one and a half orders of magnitude — while the same fluctuation is identically zero for a fixed convolutional kernel, so this measurement has no analogue there at all.

The honest partition of these results is that the *structural* claim — gauge-equivariant attention maps are lattice operators, with a mass, an overlap, and a spectral decomposition — is established, and holds already at initialisation; while the *learning* claim is established by A_0 rather than by ξ_A , and is best stated as *training removes excited-state contamination from the attention field*. The trained attention field is, by that measure, a better operator than the hand-built smeared basis at every β examined, and increasingly so towards the critical point.

What remains open is a volume: both of the largest- ξ ensembles sit at $\xi \gtrsim L/4$, so the excursion that makes the correlation persuasive is also the one feature that an $L = 32$ run could reassign to finite-volume squeezing. Until then the excursion is reported, not explained. Beyond that, the natural extensions are an entropy-matched random control to close the $\delta A/A$ confound, and the same readout applied to the $SU(2)$ glueball operator, where the machinery already exists and only the correlation length does not.